Cygnus Vertex (ATH-498)

Institutional Alternative Investment Strategy — Quantitative Performance Analysis

Track Record: Apr 2002 – Apr 2026 (289 Months / 24.1 Years) | **Benchmark:** S&P 500 Total Return

ATHÉNÉE INVESTMENT

CAPITAL INTRODUCTION & THIRD-PARTY MARKETING

CAGR (Net)	Ann. Volatility	Sharpe Ratio	Max Drawdown	Alpha (Ann.)	Beta & Corr.
10.00%	**8.96%**	**1.11**	**-11.43%**	**9.28%**	**0.06 | +0.11**
S&P 500: 9.99%	S&P 500: 14.97%	SPX: 0.71 | Sortino: 1.99	SPX: -50.95% | Calmar: 0.87	t-stat: 5.00	Beta t-stat: 1.83

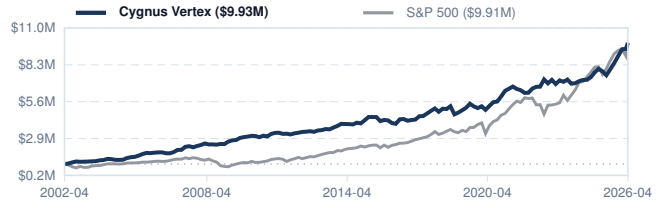
Performance, Risk & Hit Rate Indicators

Indicator	Fund	S&P 500
CAGR (Annualized)	**10.00%**	9.99%
Annualized Mean Return	9.97%	10.69%
Annualized Volatility	8.96%	14.97%
Proxy Sharpe Ratio ($r_f = 0\%$)	**1.11**	0.71
Annualized Sortino Ratio	1.99	1.08
Maximum Drawdown (MDD)	**-11.43%**	-50.95%
Calmar Ratio (CAGR / |MDD|)	0.87	0.20
Cumulative Total Return	893.29%	891.02%
Best / Worst Monthly	+8.77% / -7.56%	+12.82% / -16.80%
Positive Months (Hit Rate)	66.1% (191/289)	67.1% (194/289)
Crisis Hit Rate (Fund > 0 | SPX < 0)	**57.9% (55/95)**	–
Up-Mkt Loss (Fund < 0 | SPX > 0)	29.9% (58/194)	–

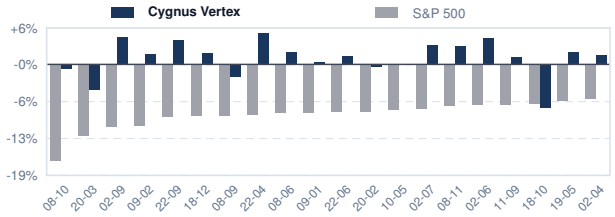
Alpha, Beta, Correlation & Tail Analysis (vs. S&P 500)

Quantitative Metric	Estimate / Diagnostic
Annualized Alpha (α) & t-stat	**9.28% (t=5.00)**
Market Beta (β_{SPX}) & t-stat	0.06 (t=1.83) ($R^2 = 1.2\%$)
Pearson Correlation (r) to S&P 500	**+0.11** (95% Blk CI: [-0.03, 0.23])
Spearman Rank Correlation (ρ)	+0.15 (95% Blk CI: [0.02, 0.27])
Upside / Downside Capture Ratio	Up: **0.34** | Down: **-0.07**
Cond. Up-Market (SPX > 0)	$\alpha = 10.4\%,\ \beta = 0.08$
Cond. Down-Market (SPX < 0)	$\alpha = -3.1\%,\ \beta = -0.14$
Tail Corr. (SPX $< Q_1$, Worst 25%)	$r = -0.11,\ \rho = -0.20\ (n = 72)$
Tail Corr. (SPX $< D_1$, Worst 10%)	$r = -0.02,\ \rho = -0.19\ (n = 29)$
Worst 20% SPX Months Avg. Return	Fund: +0.73% vs SPX: -5.57%
Cross-Asset Beta (WTI / BTC)	WTI β: 0.06 | BTC β: 0.03
Ljung-Box Serial Correlation	$p_1 = 0.50,\ p_3 = 0.57$ (Pass)

Cumulative Growth of $1M USD (Fund vs. S&P 500)



Worst 20% S&P 500 Months



Historical Monthly & Yearly Net Return Matrix (%)

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Yearly
2002	-	-	-	1.7%	2.4%	4.5%	3.4%	1.0%	4.7%	-1.8%	-0.3%	0.0%	**16.6%**
2003	0.6%	1.8%	-0.7%	-2.1%	3.7%	2.4%	-1.8%	1.4%	3.9%	1.2%	0.6%	3.5%	**15.3%**
2004	2.8%	0.6%	-2.0%	-2.5%	-1.1%	0.6%	-1.2%	1.2%	-0.2%	2.2%	7.0%	3.0%	**10.4%**
2005	1.3%	2.1%	-0.5%	2.9%	3.0%	6.2%	0.9%	0.4%	4.5%	-4.7%	4.0%	2.6%	**24.7%**
2006	-0.9%	1.0%	0.3%	1.0%	-0.8%	-2.1%	-1.3%	0.1%	-0.5%	1.3%	3.7%	5.5%	**7.3%**
2007	2.9%	-0.9%	1.2%	5.0%	5.6%	1.7%	0.1%	-7.3%	4.3%	5.0%	-0.6%	-1.9%	**15.2%**
2008	5.1%	5.3%	-0.7%	-5.4%	2.8%	2.2%	-1.8%	1.2%	-2.2%	-0.8%	3.1%	-0.8%	**7.7%**
2009	0.3%	1.8%	5.3%	1.1%	2.5%	-0.4%	1.3%	1.2%	4.2%	-1.0%	3.0%	0.6%	**21.7%**
2010	0.9%	0.7%	1.0%	-0.3%	-0.1%	-1.9%	-0.9%	4.8%	-0.9%	0.8%	-2.6%	4.4%	**5.6%**
2011	1.7%	2.0%	0.3%	2.8%	-2.3%	-3.3%	0.8%	-1.0%	1.2%	1.1%	-2.6%	1.2%	**1.7%**
2012	1.4%	2.0%	-0.7%	0.3%	1.6%	-2.6%	3.1%	1.6%	-0.5%	-1.9%	0.6%	0.0%	**4.9%**
2013	2.9%	0.3%	0.5%	0.6%	1.6%	-3.2%	2.7%	2.1%	1.6%	0.6%	2.6%	3.3%	**16.5%**
2014	0.7%	2.8%	-3.0%	-2.3%	2.2%	0.3%	-0.8%	2.0%	1.2%	-3.9%	2.6%	2.0%	**3.4%**
2015	3.8%	3.7%	1.9%	-3.9%	4.1%	-2.0%	2.0%	-5.2%	-1.6%	-0.4%	2.4%	0.4%	**4.8%**
2016	-1.2%	-3.7%	-1.0%	-0.7%	-0.7%	4.7%	3.6%	-0.3%	0.7%	0.0%	-2.8%	0.1%	**-1.5%**
2017	1.0%	2.0%	-1.2%	1.3%	4.4%	-1.0%	1.3%	4.9%	-0.7%	3.8%	0.2%	-1.7%	**15.1%**
2018	5.4%	-3.7%	-1.1%	0.2%	4.4%	-1.2%	0.7%	2.5%	1.6%	-7.6%	-4.2%	2.0%	**-1.6%**
2019	0.6%	2.0%	1.8%	1.4%	2.1%	4.4%	1.4%	1.9%	-5.4%	-2.6%	0.2%	0.9%	**8.6%**
2020	1.8%	-0.5%	-4.5%	0.0%	5.8%	2.9%	2.1%	0.8%	0.1%	0.3%	6.2%	8.8%	**25.8%**
2021	-1.6%	6.3%	-3.6%	3.1%	-0.7%	-2.3%	-0.3%	-2.2%	0.9%	0.3%	-3.5%	-0.4%	**-4.3%**
2022	0.5%	0.2%	4.7%	5.4%	-3.3%	1.4%	-1.2%	3.5%	4.2%	0.4%	-4.4%	2.5%	**14.4%**
2023	1.2%	0.5%	-4.7%	2.0%	1.9%	-1.1%	-0.4%	0.8%	1.6%	-0.1%	-4.1%	1.2%	**-1.6%**
2024	-0.7%	-0.9%	3.1%	3.4%	-2.0%	1.6%	-1.4%	0.2%	2.4%	-0.6%	5.6%	2.8%	**13.9%**
2025	0.5%	0.5%	-2.8%	-2.4%	-1.4%	3.0%	2.8%	1.6%	4.0%	5.3%	0.7%	1.7%	**14.1%**
2026	3.8%	2.4%	-1.9%	4.5%	-	-	-	-	-	-	-	-	**9.0%**

Fund Terms & Commercial Overview

Vehicle:	**Commingled Fund**	Return Basis:	**Net of Fees**	Fee Structure:	**2% Mgmt / 20% Perf**	Liquidity:	**Quarter Liquidity**
Lock-Up Period:	**No Lockup**	Min. Investment:	**1M USD**	Current AUM:	**250M**	Domicile:	**Cayman Islands**

Athénée Investment | Quai des Bergues 23, 1201 Geneva, Switzerland | arthur.dhonneur@athenee-investment.com

Capital Introduction & Third-Party Marketing Memorandum. Strictly for Institutional & Qualified Investors. Please see Page 2 for Risk & Legal Disclosures.

Risk Disclosures, Legal Notice & Quantitative Appendix

Fund Reference: Cygnus Vertex (ATH-498) (Apr 2002 - Apr 2026 (289 Months / 24.1 Years))

Important Legal Disclaimer, Risk Disclosures & Capital Introduction Notice

1. Role of Athénée Investment — Capital Introduction Services Only (No Investment Advice):

This document has been prepared by **Athénée Investment** solely for informational and discussion purposes as part of its institutional **Capital Introduction and Third-Party Marketing** activities. **This document does not constitute investment advice, financial advisory services, a personal recommendation, or an offer to sell or a solicitation of an offer to buy** any shares, units, or interests in any investment fund or financial instrument. Prospective investors must conduct their own independent due diligence and consult their own professional legal, regulatory, tax, and financial advisors before making any investment decision.

2. No Warranty of Data Accuracy or Completeness (Third-Party Manager Data):

All historical performance figures, return series, and strategy descriptions presented herein are derived from information provided directly by the respective third-party hedge fund managers or external data providers. While Athénée Investment has compiled and analyzed these quantitative metrics in good faith, **Athénée Investment has not independently audited or verified the underlying data and cannot guarantee—and makes no representation or warranty, express or implied, regarding—the accuracy, completeness, reliability, or timeliness** of any information contained in this document. Any errors, omissions, or subsequent restatements by the underlying fund manager are the sole responsibility of such manager.

3. Key Risk Factors of Alternative & Hedge Fund Strategies:

Investing in alternative investment funds and hedge funds involves a **high degree of risk** and is suitable only for sophisticated investors capable of bearing the loss of their entire investment. Material risks include, without limitation:

- **Market, Leverage & Derivative Risk:** Underlying funds may employ substantial leverage, short-selling, options, futures, swaps, and other derivative instruments, which can amplify both gains and losses and cause rapid capital erosion during extreme market dislocations.
- **Illiquidity & Lock-Up Risk:** Hedge fund interests are typically not listed on any regulated exchange and are subject to redemption gates, notice periods, lock-ups, and side-pocket arrangements.
- **Model, Regime Shift & Tail-Correlation Risk:** Quantitative and systematic strategies rely on mathematical models and historical statistical relationships (such as low equity beta or negative downside capture) that may break down or experience sudden correlation spikes during systemic liquidity crises.
- **Operational, Counterparty & Valuation Risk:** Strategies trading in over-the-counter (OTC) derivatives, digital assets, distressed debt, or emerging markets face heightened counterparty default, custody, regulatory, and mark-to-model valuation risks.

4. Past Performance & Quantitative Limitations:

Past performance is not indicative of, nor a guarantee of, future results. The value of investments and the income derived from them may go down as well as up. Simulated, bootstrap, or regression-based indicators (including Alpha, Beta, Sharpe Ratio, Sortino Ratio, and Capture Ratios) are backward-looking statistical estimates and do not represent a promise or forecast of future risk-adjusted returns.

5. Qualified Investors Only & Confidentiality:

This document is strictly confidential and is directed exclusively at **Institutional, Professional, and Qualified Investors** (including Qualified Investors within the meaning of the Swiss Collective Investment Schemes Act (CISA) and Financial Services Act (FinSA)). It may not be copied, reproduced, distributed, or disclosed to any third party, nor distributed in any jurisdiction where such distribution would be unlawful.

Appendix A: Extended Statistical, Bootstrap & Factor Diagnostics

Diagnostic Family	Methodology / Statistical Specification	Empirical Value
Pearson Correlation CI	95% Confidence Interval (IID Bootstrap vs. Moving Block $B = 3$, 1,000 iters)	Std: [-0.02, 0.22] | Blk: [-0.03, 0.23]
Spearman Rank Corr. CI	95% Confidence Interval (Rank Monotonic Co-Movement, IID vs. Block $B = 3$)	Std: [0.03, 0.26] | Blk: [0.02, 0.27]
Worst-Case Co-Dependence	97.5th Percentile Upper-Bound Block Bootstrap Correlation (PSD Projection)	$\rho_{\text{wc}} = 0.23$
Serial Correlation (Lag 1 & 3)	Ljung-Box Q-Statistic p-values testing for return smoothing / illiquidity bias	$p_1 = 0.50,\ p_3 = 0.57$ (Pass)
Conditional Up-Market OLS	Annualized Alpha & Beta estimated during positive S&P 500 months ($n = 194$)	$\alpha = 10.4\%,\ \beta = 0.08\ (t_\alpha = 2.88)$
Conditional Down-Market OLS	Annualized Alpha & Beta estimated during negative S&P 500 months ($n = 95$)	$\alpha = -3.1\%,\ \beta = -0.14\ (t_\alpha = -0.64)$
Commodity Factor (WTI)	OLS Regression against Front-Month WTI Crude Oil Monthly Returns	$\beta = 0.06\ (t = 3.02),\ \alpha = 9.2\%$
Digital Asset Factor (BTC)	OLS Regression against Bitcoin (BTC-USD) Monthly Returns	$\beta = 0.03\ (t = 1.85),\ \alpha = 8.8\%$

Appendix B: Quantitative Framework & Indicator Definitions

- **CAGR & Annualized Volatility:** Compound Annual Growth Rate is computed geometrically as CAGR $= \left(\prod_{t=1}^{T}(1 + r_t)\right)^{12/T} - 1$. Annualized Volatility is sample standard deviation scaled by $\sqrt{12}$: $\sigma_{\text{ann}} = \hat{\sigma}_m\sqrt{12}$. Proxy Sharpe Ratio assumes $r_f = 0\%$ ($\bar{r}_{\text{ann}}/\sigma_{\text{ann}}$).
- **Maximum Drawdown (MDD) & Calmar Ratio:** Maximum peak-to-trough decline of cumulative compounded wealth $W_t = \prod_{s\le t}(1 + r_s)$, defined as MDD $= \min_t\left((W_t - \max_{s\le t} W_s)/\max_{s\le t} W_s\right)$. Calmar Ratio is CAGR/|MDD|.
- **OLS Alpha (α), Beta (β) & Capture Ratios:** Estimated via Ordinary Least Squares $r_{\text{fund},t} = \alpha_m + \beta r_{\text{SPX},t} + \varepsilon_t$, where $\alpha_{\text{ann}} = 12 \times \alpha_m$. Upside (Downside) Capture measures the ratio of the fund's mean return to the S&P 500's mean return during positive (negative) S&P 500 months.
- **Tail Correlations & Moving Block Bootstrap:** Tail correlations evaluate linear (r) and rank (ρ) co-movement during S&P 500 drawdowns below the 25th percentile (Q_1) and 10th percentile (D_1). Moving Block Bootstrap ($B = 3$ months) preserves local serial dependencies when constructing 95% confidence intervals.

Athénée Investment | Quai des Bergues 23, 1201 Geneva, Switzerland | arthur.dhonneur@athenee-investment.com

© 2026 Athénée Investment. All rights reserved. For Institutional & Qualified Investors Only.